



Hadoop Cluster Notes & References

Key commands, AI assistance usage & important resources

Sections	3 — Commands, AI Usage, Resources
Hadoop Version	Hadoop 3.4.2
ZooKeeper	ZooKeeper 3.8.6
Java Runtime	OpenJDK 17
Environment	Docker Containers

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1. What I Used AI For

AI assistance was used selectively throughout this project — not as a replacement for understanding, but as a productivity tool for specific tasks.

1

Formatting Documents & PDFs

Used AI to polish documentation with consistent structure, correct grammar mistakes, and produce professionally formatted Word and PDF files — including this document.

2

Environment Files

AI helped generate clean, well-structured environment variable files (.bashrc, /etc/profile.d/ scripts, and Docker env_file configs) with correct syntax and ordering.

3

Automating Recovery After Container Failures

After losing the cluster setup 3 times due to docker-compose down, AI helped design the docker-compose.yml structure and startup scripts to automate service initialization on container restart.

4

Understanding Errors & Log Analysis

When encountering cryptic errors (like "Connection refused" during NameNode format, or SIGHUP killing daemons), AI helped interpret log output and pinpoint the root cause quickly.

5

Writing Comments for Scripts, Configs & Env Files

AI was used to write clear, accurate inline comments for shell scripts, Hadoop XML config files, and environment variable files — making the project easier to revisit and maintain.

6

Writing perfect README file on GitHub

AI was used to write clear, accurate README md file on GitHub with all setup guidance.

2. Most Important Commands

These are the essential commands used throughout the cluster setup, monitoring, and maintenance process. Each command is paired with an explanation of why it matters.



Log Inspection — HDFS & YARN Daemons

```
$ cat /opt/hadoop/logs/*namenode*.log | tail -50
```

- Shows the last 50 lines of the NameNode log. Critical for diagnosing NameNode startup failures, QJM connection issues, and failover events.

```
$ cat /opt/hadoop/logs/*datanode*.log | tail -50
```

- Inspects DataNode logs for block-reporting errors, connection timeouts to the NameNode, or disk failures.

```
$ cat /opt/hadoop/logs/*journalnode*.log | tail -50
```

- JournalNode is the backbone of HA edit-log synchronization. Checking this log is the first step when NameNode format or startup fails.

```
$ cat /opt/hadoop/logs/*zkfc*.log | tail -50
```

- ZKFC manages automatic failover via ZooKeeper. This log reveals health-check failures and fencing decisions between Active and Standby NameNodes.

```
$ cat /opt/hadoop/logs/*resourcemanager*.log | tail -50
```

- The ResourceManager log tracks job scheduling, NodeManager registrations, and YARN HA failover events.

```
$ cat /opt/hadoop/logs/*nodemanager*.log | tail -50
```

- NodeManager handles container launches on worker nodes. Useful for debugging failed MapReduce tasks or resource allocation errors.

```
$ ls /opt/hadoop/logs/
```

- Lists all log files present. Useful to confirm which daemons are running and find the exact log filename before tailing.



Docker Image Management

```
$ docker images | findstr hadoop
```

- Lists all locally saved Hadoop images. Essential for verifying that docker commit snapshots were created successfully before running docker-compose down.

```
$ docker commit node01 hadoop-node01:latest
```

- Saves the full state of node01 (all installed packages, configs, and binaries) as a new Docker image. This is the key solution to surviving docker-compose down without losing the entire Hadoop setup.

```
$ docker commit node02 hadoop-node02:latest
```

- Saves the state of node02 — the Standby NameNode and second ZooKeeper participant. Must be committed separately since each node has a unique myid for ZooKeeper.

```
$ docker commit node03 hadoop-node03:latest
```

- Saves node03 which hosts the third ZooKeeper instance and the third JournalNode alongside its DataNode/NodeManager roles.

```
$ docker commit node04 hadoop-worker:latest
```

- Saves the worker node image (node04 and node05 share the same worker role — no ZooKeeper or JournalNode, just DataNode + NodeManager).

🔍 Cluster Health Checks

```
$ hdfs haadmin -getAllServiceState
```

- Reports whether each NameNode is Active or Standby. The single most important command to verify HDFS HA is working correctly and that automatic failover is functioning.

```
$ yarn rmadmin -getAllServiceState
```

- Same as above but for YARN ResourceManagers. Confirms that one RM is Active and the other is in Standby, ensuring job scheduling continues if one RM fails.

```
$ jps
```

- Java Process Status — lists all running JVM processes on the current node. A quick sanity check to see which Hadoop/ZooKeeper daemons are actually alive (NameNode, DataNode, JournalNode, QuorumPeerMain, etc.).

3. Important Resources

The following references were essential during the research, setup, and troubleshooting phases of this project.

1

Apache Hadoop 3.4.2 — HDFS High Availability Using the Quorum Journal Manager
<https://hadoop.apache.org/docs/stable/hadoop-project-dist/hadoop-hdfs/HDFSHighAvailabilityWithQJM.html>

2

Apache Hadoop 3.4.2 — HDFS Users Guide
<https://hadoop.apache.org/docs/stable/hadoop-project-dist/hadoop-hdfs/HdfsUserGuide.html>

3

Apache Hadoop 3.3.6 Installation on Ubuntu 22.04 — Abhik Dey, Medium
<https://medium.com/@abhikdey06/apache-hadoop-3-3-6-installation-on-ubuntu-22-04-14516bceec85>

4

How to Set Up Apache Hadoop and YARN: A Step-by-Step Guide — Towards Dev

Hadoop Cluster — Notes & References	Technical Reference
	https://towardsdev.com/how-to-set-up-apache-hadoop-and-yarn-a-step-by-step-guide-8e939f1a4e44
5	Apache Hadoop Community Wiki (Confluence) https://cwiki.apache.org/confluence/display/hadoop/